

Sommerfeld representation of the electromagnetic Green's function

This set of notes is the first in a series on the conference paper [Coherent perfect absorption by a single nanoparticle](https://www.spiedigitallibrary.org/conference-proceedings-of-spie/12131/121310P/Coherent-perfect-absorption-by-a-single-nanoparticle/10.1117/12.2621039.short) (<https://www.spiedigitallibrary.org/conference-proceedings-of-spie/12131/121310P/Coherent-perfect-absorption-by-a-single-nanoparticle/10.1117/12.2621039.short>). Here, we wish to prove the Sommerfeld identity:

$$\frac{e^{i\omega r/c}}{r} = i \int_0^\infty \frac{k_\rho}{k_z} J_0(k_\rho \rho) e^{ik_z|z|} dk_\rho,$$

$$\text{where } k_z = \sqrt{(\omega/c)^2 - k_\rho^2}$$

At the outset, we note the obvious similarity between this identity and the Weyl identity (see <https://hackmd.io/@aligho/rkQCcsAOZg> (<https://hackmd.io/@aligho/rkQCcsAOZg>)). In addition, note that in the static limit (i.e. $\omega \rightarrow 0$), the Sommerfeld identity reduces to the following:

$$\frac{1}{r} = \int_0^\infty J_0(k_\rho \rho) e^{-k_\rho|z|} dk_\rho,$$

which was already shown to hold in this document:

<https://hackmd.io/@aligho/BkfSXAI0xl> (<https://hackmd.io/@aligho/BkfSXAI0xl>). For arbitrary ω , the proof is just as straightforward. We start with the Weyl identity

$$e^{ikr}/r = \frac{i}{2\pi} \int_{-\infty}^\infty \int_{-\infty}^\infty \frac{e^{iq_x x + iq_y y + i|z|\sqrt{k^2 - q_x^2 - q_y^2}}}{\sqrt{k^2 - q_x^2 - q_y^2}} dq_x dq_y$$

Writing this in cylindrical coordinates, i.e. taking $k_\rho^2 = q_x^2 + q_y^2$, gives us the following:

$$e^{ikr}/r = \frac{i}{2\pi} \int_{-\infty}^\infty \int_{-\infty}^\infty \frac{e^{ik_\rho \rho \cos(\theta) + i|z|\sqrt{k^2 - k_\rho^2}}}{\sqrt{k^2 - k_\rho^2}} k_\rho dk_\rho d\theta$$

We perform the integral over the angular variable, noting that, by the Jacobi-Anger expression, we have:

$$\int e^{iz \cos(\theta)} d\theta = 2\pi J_0(z),$$

which allows us to write:

$$e^{ikr}/r = i \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{J_0(k_\rho \rho) e^{i|z| \sqrt{k^2 - k_\rho^2}}}{\sqrt{k^2 - k_\rho^2}} k_\rho dk_\rho,$$

which is the Sommerfeld identity.